A machine-checked, uniform-in-order Abel formula for the Casoratian, in Lean 4/Mathlib

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Abstract

For k sequences $s^{(0)}, \ldots, s^{(k-1)} \colon \mathbb{N} \to R$ over a commutative ring R, all solving the order-k homogeneous linear recurrence $s(n+k) = \sum_{j=0}^{k-1} c_j(n)\, s(n+j)$, the *Casoratian* (discrete Wronskian) $C(n) = \det \left[s^{(t)}(n+i)\right]_{0 \le i,t<k}$ obeys the first-order law $C(n+1) = (-1)^{k-1} c_0(n)\, C(n)$, hence the closed product form $C(n) = \left(\prod_{l<n} (-1)^{k-1} c_0(l)\right) C(0)$. Only the *lowest* coefficient c_0 survives; $c_1, \ldots, c_{k-1}$ cancel. This is *Abel's formula* for the Casoratian — classical, going back to Casorati, the discrete analogue of Liouville's formula for the Wronskian. Our contribution is not the identity but its treatment: we give a single proof valid *uniformly in the order* k and formalize it in Lean 4/Mathlib to a clean axiom cone (a subset of {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`). The formal proof rewrites the last row of $C(n+1)$, via the recurrence, into a linear combination of the *cyclically rotated* rows of $C(n)$; `Matrix.det_updateRow_sum` collapses the combination to the single coefficient $c_0(n)$ and `sign_finRotate` supplies the cyclic sign $(-1)^{k-1}$. We also formalize the constant-coefficient power form $C(n) = \left((-1)^{k-1} c_0\right)^n C(0)$, an integral-domain non-vanishing (linear-independence) certificate, and the order-2 and order-3 instances proved independently by hand as faithfulness witnesses (order 2 reproduces the corpus sign $-c_0$, with Euler's $(-1)^n$ convergent Wronskian as the case $c_0 \equiv 1$; order 3 gives $+c_0$). The result opens the order-≥ 3 direction in a formalized polynomial-continued-fraction corpus previously confined to the order-2 (three-term) case — the algebraic backbone of simultaneous / Hermite–Padé approximation and of Apéry-type irrationality certificates. An independent computer-algebra gate confirms the identity symbolically at $k = 3$, by exact-rational checks for $k = 2, \ldots, 7$, and on named instances.

2020 MSC: 39A06 (primary), 68V20, 11J72, 15A15. **Keywords:** Casoratian, discrete Wronskian, Abel's formula, linear difference equation, higher-order recurrence, Hermite–Padé approximation, Apéry-type irrationality, Lean 4, Mathlib, formal verification, polynomial continued fractions.

1 Introduction

The convergents of a polynomial continued fraction $K_{n \ge 1}\, a_n/b_n$ are governed by a three-term recurrence $q_n = b_n q_{n-1} + a_n q_{n-2}$, and the basic structural invariant is the order-2 Casoratian (discrete Wronskian) of two solutions: it flips sign and scales by the lower coefficient at each step, so the convergent Wronskian $p_{n+1} q_n - p_n q_{n+1} = (-1)^n$ is constant up to sign. This is the certificate behind Euler's irrationality criterion, and across the deposited corpus the Casoratian appears only in this order-2, three-term form ([12, 11]; formalized as `casoratian_eqG` in the companion `pcf-delta` project [13]).

The order-≥ 3 direction. The natural next object is the Casoratian of $k \geq 3$ solutions of an order-k recurrence

$$s(n+k) = \sum_{j=0}^{k-1} c_j(n)\, s(n+j), \qquad c_0, \ldots, c_{k-1} \colon \mathbb{N} \to R, \tag{1}$$

the $k \times k$ determinant

$$C(n) = \det \left[s^{(t)}(n+i)\right]_{0 \leq i,t < k}. \tag{2}$$

This is the algebraic backbone of *simultaneous* (Hermite–Padé) approximation, the engine of Apéry-type irrationality proofs ([4]): there one builds several sequences satisfying one common higher-order recurrence and certifies their linear independence through the non-vanishing of exactly this determinant. The first-order law that C obeys is classical — the discrete analogue of Abel's identity for the Wronskian of a linear ODE, going back to Casorati; see, e.g., [1, Ch. 2] and [2]. It states that C is reproduced at each step by a single scalar built only from the *lowest* coefficient c_0.

Contribution. The identity itself is not new. What we provide is:

(i) a single proof valid *uniformly in the order k* (Theorem 1), and its formalization in Lean 4 with Mathlib to a clean axiom cone (Section 5), so the result is PROVEN in the strict sense of this program: `#print axioms` reports a subset of {`propext`, `Classical.choice`, `Quot.sound`} with no `sorryAx` and no `sorry`/`admit`/`native_decide` in source;

(ii) the closed product form, the constant-coefficient power form, and an integral-domain non-vanishing (linear-independence) certificate, all formalized (Section 4);

(iii) the order-2 and order-3 instances proved *independently* by a direct cofactor computation, kept as faithfulness witnesses against the general proof (Proposition 1); and

(iv) an independent computer-algebra gate (Section 6) that re-derives the law symbolically and on named instances, in the program's "gate-before-Lean" discipline.

Concretely, this opens the order-≥ 3 space inside a formalized PCF corpus that had treated only order 2, and supplies a reusable, machine-checked independence certificate for future Hermite–Padé constructions.

2 The Casoratian of a linear recurrence

Throughout, R is a commutative ring and $k \geq 1$ a fixed order. We write a solution of (1) simply as a sequence $s \colon \mathbb{N} \to R$ satisfying that identity for all n; no invertibility of the coefficients is assumed.

Definition 1 (Casoratian)**.** *For sequences $s^{(0)}, \ldots, s^{(k-1)} \colon \mathbb{N} \to R$, the* Casoratian *at n is the determinant* (2) *of the $k \times k$ matrix whose (i,t) entry is the i-shift of the t-th sequence, $s^{(t)}(n+i)$.*

Because det is invariant under transposition, it is immaterial whether one indexes rows by shifts and columns by solutions or vice versa; we fix rows = shifts. The two "ends" of $C(n+1)$ are what drive the law:

- **Top $k-1$ rows.** For $0 \leq i \leq k-2$, row i of $C(n+1)$ is the shift vector $\left(s^{(t)}(n+1+i)\right)_t$, which is precisely row $i+1$ of $C(n)$. So the top $k-1$ rows of $C(n+1)$ are the bottom $k-1$ rows of $C(n)$, in order.
- **Bottom row.** Row $k-1$ of $C(n+1)$ is the shift vector at $n+k$, $\left(s^{(t)}(n+k)\right)_t$, which by the recurrence (1) equals $\sum_{j=0}^{k-1} c_j(n) \left(s^{(t)}(n+j)\right)_t$, a linear combination of *all* k rows of $C(n)$.

3 The general-order Abel formula for the Casoratian

Theorem 1 (Abel's formula for the Casoratian, uniform in k)**.** *Let $s^{(0)}, \dots, s^{(k-1)} : \mathbb{N} \to R$ all satisfy the order-k recurrence* (1)*. Then the Casoratian* (2) *satisfies the first-order law*

$$C(n+1) = (-1)^{k-1}\, c_0(n)\, C(n) \qquad (n \in \mathbb{N}), \tag{3}$$

and consequently the closed product form

$$C(n) = \left(\prod_{l=0}^{n-1} (-1)^{k-1} c_0(l)\right) C(0) = (-1)^{(k-1)n} \left(\prod_{l=0}^{n-1} c_0(l)\right) C(0). \tag{4}$$

Only the lowest coefficient c_0 enters; $c_1, \dots, c_{k-1}$ cancel.

Proof. Let ρ be the cyclic permutation of $\{0, 1, \dots, k-1\}$ that moves every index up by one and wraps the top to the bottom, $\rho(i) = i + 1$ for $i < k - 1$ and $\rho(k-1) = 0$; it is a single k-cycle, so $\operatorname{sgn} \rho = (-1)^{k-1}$. Let M be $C(n)$ with its rows permuted by ρ, i.e. row i of M is row $\rho(i)$ of $C(n)$. By the observation in Section 2, the top $k-1$ rows of M (namely rows $1, \dots, k-1$ of $C(n)$) are exactly the top $k-1$ rows of $C(n+1)$, while the bottom row of M is row 0 of $C(n)$.

Replacing the bottom row of M by the linear combination $\sum_{j=0}^{k-1} c_j(n) \big(\text{row } j \text{ of } C(n)\big)$ produces exactly $C(n+1)$, because that combination is the bottom row of $C(n+1)$ and the top rows already agree. Now take determinants. Multilinearity in the replaced row — the content of `Matrix.det_updateRow_sum` — gives

$$\det C(n+1) = \sum_{j=0}^{k-1} c_j(n)\, \det\big(M \text{ with bottom row } = \text{row } j \text{ of } C(n)\big).$$

Every summand with $j \geq 1$ places into the bottom position a row of $C(n)$ that already occurs among the top $k-1$ rows of M, so that determinant vanishes. Only $j = 0$ survives, and there the bottom row is the original row 0 of $C(n)$, so the summand is $c_0(n) \det M$. Finally $\det M = \operatorname{sgn} \rho \cdot \det C(n) = (-1)^{k-1} \det C(n)$. Combining gives (3). A trivial induction on n, using $C(0)$ as the base case and (3) at the inductive step, yields the product form (4). □

Remark 1 (What the formalization does, verbatim)**.** *The Lean development (Section 5) realizes this proof literally. It builds M as* `(casoMat s n).submatrix (finRotate (k+1)) id` *(Mathlib's* `finRotate` *is the cycle ρ), expresses $C(n+1)$ as that submatrix with its last row updated to the recurrence combination, then rewrites with* `Matrix.det_updateRow_sum` *(the multilinear collapse to the surviving coefficient) and* `Matrix.det_permute` *together with* `sign_finRotate` *(the cyclic sign). The "vanishing for $j \geq 1$" step is automatic: it is exactly the statement that* `det_updateRow_sum` *isolates the coefficient indexed by the updated row. No analytic input is used; the argument is finitary algebra over an arbitrary* `CommRing`*.*

4 Specializations and consequences

Corollary 1 (Constant lowest coefficient)**.** *If $c_0(l) = a$ for all l, then $C(n) = \big((-1)^{k-1}a\big)^n C(0)$. In particular, if a is a unit (or 1), the Casoratian never vanishes once $C(0) \neq 0$.*

Corollary 2 (Non-vanishing / linear-independence certificate)**.** *Suppose R is an integral domain. If $C(0) \neq 0$ and $c_0(l) \neq 0$ for every $l < n$, then $C(n) \neq 0$. Hence if $C(0) \neq 0$ and c_0 is nowhere zero, the k solutions stay linearly independent at every window: $C(n) \neq 0$ for all n.*

Proof. Immediate from (4): a product of nonzero elements of a domain is nonzero, and $(-1)^{k-1}$ is a unit. $\square$

Corollary 2 is the order-k analogue of the order-2 statement $p_{n+1}q_n - p_n q_{n+1} = (-1)^n \neq 0$ that underlies Euler's criterion: it is precisely the determinantal independence certificate one needs for Hermite–Padé systems.

Proposition 1 (Low-order witnesses)**.** *The cases $k = 2, 3$ hold and are proved independently by direct cofactor expansion, without the permutation argument:*

$$k = 2: \quad C(n+1) = -\,c_0(n)\,C(n); \qquad k = 3: \quad C(n+1) = +\,c_0(n)\,C(n).$$

The order-2 sign $-c_0$ is the corpus sign; its $c_0 \equiv 1$ instance is the constant convergent Wronskian $p_{n+1}q_n - p_n q_{n+1} = (-1)^n$.

These hand proofs (`caso2_step`, `caso3_step`; the 3×3 cofactor expansion is closed by `ring`) are retained as *faithfulness witnesses*: they agree with Theorem 1 at $k = 2, 3$, guarding against a vacuous or mis-signed general statement.

5 The machine-checked formalization

The development is in Lean 4 with Mathlib (toolchain pinned `leanprover/lean4:v4.30.0`, Mathlib `rev=v4.30.0`). It is built and axiom-checked inside the companion `pcf-delta PcfContinuant` project [13], whose `Check.lean` gate runs `#print axioms` on every declaration; the two source files `GeneralCaso.lean` (the uniform-in-k core, namespace `PcfGeneralCaso`) and `HigherCaso.lean` (the by-hand low-order witnesses, namespace `PcfHigherCaso`) are mirrored in this catalogue under `verify/`.

Remark 2 (Index convention)**.** *The formalization parametrizes the order as $k+1$ via `Fin (k+1)` (so the recurrence is $s(n + (k+1)) = \sum_{j \le k} c_j(n) s(n+j)$ and the matrix is $(k+1) \times (k+1)$); its sign therefore reads $(-1)^k$. This is the prose order-k law* (3) *under $k \mapsto k+1$: an index shift only, with $(-1)^{(k+1)-1} = (-1)^k$.*

Theorem 2 (Formalized core; PROVEN)**.** *The following nine declarations are machine-checked with clean axiom cones — each a subset of $\{$`propext`, `Classical.choice`, `Quot.sound`$\}$, no `sorryAx`, sources free of `sorry`/`admit`/`native_decide`, `lake build` exit* 0.

- Uniform in k *(`PcfGeneralCaso`, `GeneralCaso.lean`): `casoMat_det_step` is the step law* (3) *$C(n+1) = (-1)^k c_0(n) C(n)$ (order $k+1$; Remark 2); `casoMat_det_eq` is the closed product form* (4)*; `casoMat_det_eq_const` is the constant-coefficient power form (Corollary 1); `casoMat_det_ne_zero_of_init` is the integral-domain non-vanishing certificate (Corollary 2).*
- Low-order witnesses *(`PcfHigherCaso`, `HigherCaso.lean`): `caso2_step` (the order-*2 *sign $-c_0$), `caso3_step` (the order-*3 *step $+c_0$), `caso3_eq` (its product form), `caso3_eq_const` (its constant-coefficient case), and `caso3_ne_zero_of_init` (its domain non-vanishing certificate) — Proposition 1.*

Of the nine, exactly two close with the minimal cone $\{$`propext`, `Quot.sound`$\}$ — the by-hand witnesses `caso2_step` and `caso3_step`; the remaining seven additionally use `Classical.choice`. This `Classical.choice` usage is inherited from Mathlib's general determinant infrastructure (`Matrix.det` and the underlying permutation machinery), not a sign that the uniform-in-k proof is in any meaningful sense less constructive than the by-hand cases; we report the cones factually rather than rank them.

This follows the program's *unconditional-core* pattern: there is no labelled analytic hypothesis here at all, since the statement is pure finitary algebra over a `CommRing` (the domain results add only `IsDomain R`). The Casoratian is defined as a genuine `Matrix` determinant, so the result is the honest determinantal identity, not a bespoke $k = 2, 3$ polynomial expression. Only the order-2 *step* sign is recorded here, as `caso2_step`: the full order-2 closed-form suite already exists in `pcf-delta` as `casoratian_eqG` [13] and is not duplicated, whereas order 3 carries the full suite as the smallest genuinely higher-order witness.

6 Independent symbolic verification

In keeping with the catalogue's gate-before-Lean discipline, a `sympy` harness (`harness_caso_k.py`) re-derives the law independently of the Lean proof:

- a fully *symbolic* proof at $k = 3$: with generic symbolic data and generic c_0, c_1, c_2, $\det C(1) - c_0 \det C(0)$ expands to 0 and $\det C(1)$ is free of c_1, c_2 (only c_0 enters);
- *exact-rational* checks of both the step law (3) and the product form (4) for $k = 2, \ldots, 7$, over integer-coefficient recurrences and initial data drawn from a *fixed deterministic seed* (`random.Random(7k+101t+5)` for trial t), so the runs are exactly reproducible, confirming the alternating sign $(-1)^{k-1}$; and
- named instances: the Tribonacci recurrence $s(n+3) = s(n+2) + s(n+1) + s(n)$ ($c_0 \equiv 1$) has a *constant* Casoratian, and the recurrence with $c_0(n) = n + 1$ gives $C(n) = n!\, C(0)$ — both matching (4) exactly.

All checks pass. The harness and the Lean sources are listed in Section 9.

7 Epistemic status

- **PROVEN** (Lean, clean cones; Theorem 2): the uniform-in-k step law and product form, the constant-coefficient power form, the integral-domain non-vanishing certificate, and the order-2/order-3 witnesses. No analytic input; the cones are subsets of {`propext`, `Classical.choice`, `Quot.sound`} with no `sorryAx`.
- **VERIFIED** (numerically/symbolically; Section 6): the law at $k = 3$ symbolically and for $k = 2, \ldots, 7$ by exact-rational checks, plus the named instances.
- **STRUCTURAL** (classical; formalization pending): the *fundamental-system converse* — that when $C(n_0)$ is a unit (over a domain, with c_0 invertible) every solution of (1) is an R-linear combination of the $s^{(t)}$. This is the classical fundamental-set theorem ([1, Ch. 2]); over a field it is immediate from the solution space being pinned by the first k values, as Mathlib's `LinearRecurrence` already encodes in the constant-coefficient case [6]. Over a general commutative ring it is less automatic, and two *distinct* hypotheses enter: one needs c_0 to be a unit (so the recurrence is reversible and its solution set is a free R-module of rank k, pinned by any k consecutive values), and one needs $C(n_0)$ to be a unit (so the chosen solutions $s^{(0)}, \ldots, s^{(k-1)}$ actually span that module at the window n_0). The two interact through the step law (3) — with c_0 a unit at every step, C is a unit at one window iff at all — but neither implies the other. This is exactly why we record the converse here as STRUCTURAL rather than asserting it in that generality. We have not yet formalized it in the present uniform-in-k, time-varying setting.

- **CONJECTURED / open**: a concrete simultaneous-Hermite–Padé PCF whose order-3 Casoratian, via Corollary 2, furnishes an Apéry-style irrationality certificate. This is the genuinely open next entry; the present note — together with the converse above — supplies the determinantal machinery it will rely on.

We claim no novelty for Abel's formula itself, which is classical ([1, Ch. 2],[2],[3]); the contribution is the uniform-in-k machine-checked proof and the opening of the order-≥ 3 direction in this corpus.

8 Relation to prior work and novelty

Formal-methods landscape. The two standard formal-library treatments of linear recurrences both target the *constant-coefficient* solution theory, not the Casoratian evolution law. Mathlib's `LinearRecurrence` [6] fixes an order and a tuple of *constant* coefficients `coeffs : Fin order → R`, builds the solution submodule of $\mathbb{N} \to R$, the `LinearEquiv` sending a solution to its first `order` terms (so solutions are pinned by their initial data), and the characteristic polynomial; it defines no Casoratian and proves no Abel-type evolution law. Eberl's *Linear Recurrences* entry in the Isabelle Archive of Formal Proofs [9] develops constant-coefficient recurrences via formal power series and polynomial factorisation into an executable closed-form solver, again without the Casoratian/Abel identity. Our object is complementary and, in one direction, strictly more general: the coefficients $c_j(n)$ are arbitrary functions of n (time-varying), the Casoratian is a genuine `Matrix` determinant, and the theorem is the determinantal evolution law uniform in the order k — exactly the non-vanishing / linear-independence certificate (Corollary 2) that Hermite–Padé and Apéry-type constructions consume ([4, 5]). A machine-checked Casoratian does already exist for the order-2 case in another system: the Coq/MathComp formal proof of the irrationality of $\zeta(3)$ [10] defines the 2×2 Wronskian `ba_casoratian` of the two Apéry sequences and derives its first-order (Abel) law, but only for that concrete instance — indeed its source remarks that the order-1 recurrence "should be obtained from something more general." Having searched Mathlib, the Isabelle AFP, and the Coq/MathComp ecosystem by name (*Casoratian*, *Wronskian*) and by structure (as of June 2026), we find no *machine-checked* general-order, uniform-in-k Casoratian Abel formula with time-varying coefficients in any of these systems — precisely the "something more general" the Apéry formalization lacked. The *identity* itself is of course classical and not in question: the general-order law with time-varying coefficients is textbook ([1, Ch. 2],[2],[3]), for which we claim no novelty. Our claim is strictly one of *formalization coverage* — the machine-checked theorem, uniform in k and with time-varying $c_j(n)$, that the present note contributes and gate-checks.

Within this corpus, and a note on hosting. The deposited corpus had formalized the Casoratian only in its order-2 form — the $4/\pi$ PCF Casoratian closed form (Theorem 6.3 of the ladder paper [12]; machine-verified in [11]) and the general order-2 `casoratian_eqG` in `pcf-delta` [13] — so this is the program's first order-≥ 3 result and the backbone of a planned Hermite–Padé / Apéry line. We deposit it as a standalone catalogue entry rather than as a Mathlib pull request for two reasons: the result is more general than Mathlib's constant-coefficient `LinearRecurrence` API and would need new time-varying-coefficient scaffolding to land there, and the corpus requires the core to be hosted and cone-gated inside `pcf-delta`'s `Check.lean`. A Mathlib contribution — a `Casoratian` determinant and its Abel evolution law, specialising to `LinearRecurrence` in the constant-coefficient case — is a natural next step and would convert "not otherwise available" into an externally reviewed library lemma. For the record, the Lean and `sympy` sources label this identity "Abel–Jacobi–Liouville"; the standard discrete name, used here, is Abel's formula (Elaydi [1]), the triple form naming the continuous-Wronskian analogue (Liouville / Jacobi). The note supplements the SIARC program [14].

9 Code and data availability

The Lean sources are `verify/GeneralCaso.lean` (namespace `PcfGeneralCaso`) and `verify/HigherCaso.lean` (namespace `PcfHigherCaso`); they are built and axiom-cone-checked inside the `pcf-delta PcfContinuant` project [13] via `lake env lean Check.lean` (toolchain `leanprover/lean4:v4.30.0`, Mathlib `rev=v4.30.0`); the verbatim `#print axioms` output for the nine declarations is mirrored in this repository as `verify/AXIOM_CONES.txt`, so the cone partition of Section 5 is reproducible here without rebuilding the host project. The independent symbolic gate is `harness_caso_k.py` (`sympy` [8]; run `python harness_caso_k.py`), which prints the $k = 3$ symbolic proof, the exact-rational $k = 2, \ldots, 7$ verification, and the named instances. Public repository: `https://github.com/papanokechi/pcf-casoratian-catalogue`.

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